

Fusion+ : A Dual-CNN Hybrid Framework for Rapid Post-Disaster Building Damage Assessment

Computer Vision Research Project

Kaggle Notebook: <https://www.kaggle.com/code/saditarshad/fusionplus-disaster-assessment>

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Abstract

Rapid and accurate building damage assessment following natural disasters is critical for humanitarian response and resource allocation. However, because models lack a structural baseline, it is very challenging to determine precise damage categories using post-disaster photos alone. To bridge this gap, we developed the Fusion+ Hybrid Framework, a Siamese-inspired dual-stream CNN that simultaneously evaluates pre and post-disaster satellite pairs from the xView2 (xBD) dataset. Stream A generates a 2,048-dimensional vector by mapping pre-existing structural geometry using a frozen ResNet50 architecture. In the meantime, Stream B generates a 1,280-dimensional vector by isolating post-disaster debris and texture anomalies using a frozen EfficientNet-B0. These representations are merged into a 3,328-dimensional fusion vector and processed by a custom MLP bottleneck classifier. Because the raw data suffers from a severe long-tail imbalance 75% of samples show 'No Damage' and only 7% are 'Destroyed' we aggressively under-sampled dominant classes to construct a balanced training set of 40,000 samples. Training exclusively on the lightweight classification head allows the network to converge in fewer than 30 epochs using early stopping. When evaluated on an 8,000-sample validation set, the framework achieved a 77% Macro F1-Score and 77% overall accuracy across four tiers: No Damage, Minor Damage, Major Damage, and Destroyed. Ultimately, these results prove that dual-stream feature fusion paired with frozen backbones yields a highly efficient, robust model capable of running directly on mobile CPU hardware.

1. Introduction

1.1 Background and Motivation

Each year a lot of people lose their homes and many buildings are. Natural disasters like hurricanes, earthquakes, floods, wild fires, volcanic eruptions etc. causes a lot of problems and also leads to a lot of economic damage. A quick look at the damage is an essential component to assisting people and working together and to deal with disasters such as hurricanes, earthquakes and floods and wildfires and volcanic eruptions. It is not practical or cost effective to do a site survey to assess the damage. May require a considerable amount of time in difficult to access or severely damaged areas.

Satellite images are very helpful now because they are very clear. Satellite images are a help in figuring out what happened after a big disaster. With technology Satellite images are available and they can help us see what is going on from far away. Satellite platforms can respond quickly to a disaster area by giving us the information without having to go into the disaster area. This helps us understand what is happening in the disaster area. However it is still hard to recognize buildings from satellite images because buildings are different and the environment is different and the disaster is different. Recognizing buildings from satellite images of disasters, like hurricanes and earthquakes and floods and wild fires and volcanic eruptions is still a tough problem. The use of only post disaster images is a significant constraint of many traditional methods. Single image approaches often do not identify real damage caused by disaster conditions from pre-existing structural irregularities or poor-quality infrastructure. Such approaches tend to lack classification reliability and ambiguity in context.

In recent research, multi-temporal analysis with the help of paired pre- and post-disaster images has received a lot of focus, as a way to overcome this limitation. Pre-disaster images contain baseline structural information, post-disaster images contain debris patterns, roof collapse, flooding effect and other anomalies related to the damage. All these complementary features allow more precise and contextual damage classification.

The xView2 challenge dataset was created from a benchmark dataset (xBD) where a large number of pre and post disaster satellite images were georeferenced and annotated with a universal four tier damage classification scale with values No Damage, Minor Damage, Major Damage, and Destroyed. The dataset also includes real world difficulties including extreme class imbalance as well as degree of similarity in damage classes for the images.

Inspired by these challenges, this research suggests a dual-stream deep feature fusion network for automated disaster damage assessment, called Fusion+. The proposed model combines the ResNet50-based structural feature extraction and the EfficientNetB0-based textural analysis, to learn the spatial and the disaster-specific representations jointly from multi-temporal satellite images. Furthermore, the framework integrates imbalance-aware preprocessing and lightweight transfer learning methods to attain computational efficiency and high classification accuracy appropriate for the application in humanitarian tasks in the field.

1.2 Core Machine Learning Challenges

Using satellite imagery to automatically assess disaster damage has several important machine learning problems that have direct impacts on the reliability of the model and its generalization. Of these two, two are especially important: extreme class imbalance and high inter-class visual ambiguity.

The first challenge is related to the highly skewed distribution of disasters in the real world. The xBD/xView2 dataset contains about 75% instances in the No Damage class and almost 7% in the Destroyed class. These long-tail class distributions make deep learning models focus on the majority classes while training, and may lead to failure in recognizing severely damaged structures. However, the

situation changes when comparing conventional evaluation result measures like overall accuracy. A simple "no damage" classifier, for instance, might be accurate for the majority of buildings, without being able to identify the ones seriously damaged, which are the most crucial targets for humanitarian response and emergency planning. Thus, it is essential that the imbalance-aware preprocessing and evaluation strategies are used to ensure good classification of disaster.

A second challenge is the high visual overlap of the damage categories, adjacent to each other such as Minor Damage and Major Damage. When viewed from an orbital nadir, subtle features like partial roof collapse, facade deformation or local debris can be hard to see because of the low resolution of satellite images. This gives rise to non-separability of feature distributions over intermediate damage classes and consequent high rates of misclassification. This visual ambiguity poses a great challenge for building a reliable multi-class damage prediction system in practical disaster scenes.

1.3 Proposed Solution and Contributions

This paper introduces the Fusion+ hybrid Framework, which makes the following key contributions:

- A dual-stream Siamese-inspired CNN architecture processing pre/post image pairs simultaneously.
- Strategic frozen transfer learning from ImageNet-pretrained ResNet50 and EfficientNet-B0, enabling training in minutes on consumer-grade hardware.
- A class-balanced extraction pipeline that resolves long-tail imbalance without synthetic data generation.
- Achieving 77% Macro F1-Score on a held-out balanced 8,000-sample validation set.

2. Literature Review

2.1 Traditional Change Detection Methods

The first attempts to assess damages from satellites were based on the difference in spectral values between pre- and post-disaster images at the pixel level. Per-pixel feature differences are computed using change vector analysis (CVA) or multivariate alteration detection (MAD) and then some threshold-based classification is applied. Although simple to compute, they are very sensitive to atmospheric conditions, variations of illumination, and sensor differences between acquisition dates, resulting in a high number of false positives in areas of non-structural change, e.g. vegetation regrowth or seasonal change.

To put these limitations in perspective, Dong and Shan (2013) presented a basic classification of remote sensing damage detection methods into multi-temporal methods and mono-temporal methods. Although theoretically desirable, multi-temporal approaches were not always feasible in practice as they involved processing of heterogeneous data sets under hasty timelines in a crisis situation. In addition, Dong and Shan found there is a significant spatial resolution limitation: damage only can be quantified at the neighbourhood or block level for resolution 10 m and above. Specific, individual structural degradation

will need meter, sub-meter visual fidelity. Due to this lack of high resolution, standardized pre-event baselines, many operational workflows used in the early days had to settle for mono-temporal analysis. This post-event approach has been applied to identify the structural signatures (texturing, morphological shape variations, and structural shadows) which is a difficult task in the absence of a comparative baseline to assign precise damage grades. Before the era of deep learning, early literature focused on complementary uses of Optical imagery and high-resolution shape extraction, LiDAR and 3D volumetric and elevation changes, and Synthetic Aperture Radar (SAR) and all-weather phase coherence while systematically fusing these multimodal features was an unsolved challenge.

2.2 Deep Learning in Damage Mapping

Convolutional Neural Networks (CNNs) have progressively replaced classical methods for remote sensing change detection. Early deep learning approaches applied single-stream CNNs exclusively to post-disaster imagery, framing damage assessment as a standard image classification problem. A comprehensive meta-analysis of 242 papers by Wang et al. (2024) reveals that these single-temporal, post-event pipelines still dominate 66% of the literature. This dominance is heavily driven by their operational agility during rapid emergency response, as they bypass the need for intense data preprocessing and exact image registration required by historical baselines. While this improved over classical pixel-differencing, the absence of pre-disaster structural baseline context systematically limited classifier performance, particularly in distinguishing complex urban damage from pre-existing natural debris.

Subsequent research explored Siamese network architectures that share weights across two identical streams processing pre- and post-disaster images respectively, computing a learned difference representation. This bitemporal shift was historically delayed by a massive data bottleneck; as Wang et al. (2024) noted, prior to 2019, only 13 public datasets were available globally, severely capping deep learning scalability. This paradigm shifted dramatically with the introduction of the xBD dataset (Gupta et al., 2019). By providing over 850,000 sub-meter ($<0.8\text{m}$ Ground Sample Distance) building polygons across 19 sudden-onset global disasters, xBD established the largest structural diversity benchmark in the field. Furthermore, it solved the historical reliance on disjointed, binary ("damaged" vs. "undamaged") scoring by introducing a standardized, expert-validated 4-level Joint Damage Scale: No Damage, Minor Damage, Major Damage, and Destroyed. Gupta et al. also established the "negative imagery imperative," proving that algorithms require a heavy quota of non-damaged buffer imagery to statistically separate natural environmental baselines from disaster impact.

The subsequent xView2 competition established state-of-the-art benchmarks utilizing U-Net and DeepLabv3+ based segmentation architectures for pixel-level damage mapping, successfully moving the field from coarse bounding boxes to exact structural footprint extraction.

2.3 Generalization Roadblocks and Domain Adaptation

Despite the availability of expansive datasets like xBD, scaling these deep learning architectures to real-world deployment faces severe out-of-distribution (OOD) vulnerabilities. Every disaster is environmentally and architecturally unique. Wang et al. (2024) demonstrated that while modern CNNs and Deconvolutional Networks can achieve up to 95.82% accuracy in binary classification, performance plummets to an incredibly low 14% to 86.5% range when scaled to the granular 4-level xBD scale. Models fundamentally struggle to differentiate between "minor" and "major" structural damage, an issue heavily exacerbated by "rooftop vs. facade blindspots," where top-down nadir satellite imagery naturally misses foundational wall cracks visible only from ground or UAV perspectives.

To combat this OOD generalization failure without the impossible task of manually labeling post-disaster imagery during a live crisis, contemporary research has pivoted to generative artificial intelligence. Kalluri et al. (2024) pioneered a zero-shot target framework utilizing a pre-trained, non-autoregressive text-to-image transformer (MUSE). Instead of relying on real post-disaster data, their pipeline passes a target domain's pre-disaster image through a VQGAN encoder, applies a localized binary mask over the structure, and uses cross-attention text prompts to synthesize high-fidelity, highly contextual post-disaster destruction (e.g., simulating flood water or wind damage). By fine-tuning models on this synthetically generated target data, Kalluri et al. raised cross-geographical multi-source transfer accuracy from a collapsed baseline of 25.27% up to 54.40%, an absolute improvement of +29.13%.

2.4 Operational Implementation and the Fusion+ Framework

While generative AI can be used as an offline capability, active disaster response requires localized strategies at run-time. As brought to light by tornadoes in 2023 in Rolling Fork, Mississippi, researchers moved from pre-trained algorithms to a Human-in-the-Loop (HITL) approach, in conjunction with the American Red Cross. The workflow enabled human analysts to quickly annotate the disaster imagery acquired with localized context-specific models trained on the scene. It helped to overcome the data-diversity issues which can pose a major challenge to pre-trained models, attaining an outstanding precision of 0.86 and recall of 0.80. Importantly, the entire end-to-end processing chain was performed within a few hours per satellite scene, with an impressive human adaptability to do the same for real-time processing in many cases where the historical pre-training is not possible.

These state-of-the-art bitemporal segmentation approaches and active learning pipelines, however, have high computational requirements and need expensive GPUs to run their inference. This structural need is extremely restrictive for use in the field, especially for localized responders in low resource/power areas during immediate post event operations.

The Fusion+ Framework is designed to overcome the limitations of the single-stream classifiers and segmentation models in the identified tasks by combining the two streams using a lightweight classification head, which extracts features from the structural baseline geometry and damages signatures extracted from the image texture.

3. Methodology: The Fusion+ Framework

3.1 Architectural Pipeline Overview

The proposed Fusion+ framework is a deep feature fusion architecture with two streams that are designed for the automated satellite-based disaster damage assessment. The model processes building image crops in pairs from pre-disaster and post-disaster sites to obtain pre-disaster structural characteristics in addition to post-disaster unexpected changes. To localize each building instance, polygon annotations from xView2 are used, and then they are resized to a standard spatial resolution of 224×224 pixels for consistency with the pretrained convolutional neural network (CNN) backbones used for feature extraction.

The architectural pipeline is a parallel processing stream of feature extraction processes. The first stream is used to extract high-level structural and geometric features from pre-disaster images using a pretrained ResNet50 network, and the second stream is used to extract fine-grained features like debris patterns, roof collapse and surface irregularities from the post-disaster images using EfficientNetB0 network. Both backbone networks are frozen to save computational resources and keep the generalized ImageNet representations.

Once the features are extracted, the output vectors from both streams are combined to make a single high dimensional representation with complementary spatial and textural information. The fused feature vector then goes through a trainable multi-layer perceptron (MLP) classification head, which includes nonlinear activation layers, dropout regularization and a bottleneck reduction. Finally, there is an output layer based on Softmax which outputs a probability for each one of the four categories of disaster damage, No Damage, Minor Damage, Major Damage and Destroyed.

The overall pipeline is aimed to have a balance between computational efficiency, feature diversity, and strong multi-class disaster classification performance. The entire architectural pathway is summarized below:

INPUT LAYER

Pre-Disaster Image ($224 \times 224 \times 3$) + Post-Disaster Image ($224 \times 224 \times 3$)

STREAM A — ResNet50 (Frozen)

Pre-Disaster Structural Features → Output: 2,048-dim vector | Captures: footprint geometry, spatial hierarchy

STREAM B — EfficientNet-B0 (Frozen)

Post-Disaster Textural Features → Output: 1,280-dim vector | Captures: debris patterns, scorch marks, surface anomalies

FEATURE FUSION LAYER

Vector Concatenation: $[2,048 \parallel 1,280] = 3,328$ -dimensional fused representation

NEURAL BOTTLENECK MLP HEAD (Trainable)

$3,328 \rightarrow \text{FC}(512) + \text{BN} + \text{ReLU} + \text{Dropout}(0.4) \rightarrow \text{FC}(256) + \text{BN} + \text{ReLU} + \text{Dropout}(0.3) \rightarrow \text{FC}(128) + \text{BN} + \text{ReLU} \rightarrow \text{FC}(4) + \text{Softmax}$

OUTPUT

4-Class Softmax: { No Damage | Minor Damage | Major Damage | Destroyed }

Figure 1. Fusion+ Dual-Stream CNN Architecture. Frozen pre-trained backbones extract complementary structural and textural features, fused through a trainable bottleneck classifier.

3.2 Stream A: Pre-Disaster Spatial Geometry (ResNet50)

To extract structure and geometry from pre-disaster satellite images, with a pretrained ResNet-50 backbone, Stream A is tasked with doing so. The general goal of this stream is to collect a stable baseline characterization of building footprint geometry, architectural consistency and structural completeness before the disaster event. Pre-disaster imagery is undamaged, thus bearing critical information of buildings, which is used as a reference to identify the changes that happened after the disaster.

ResNet50 was chosen for its powerful residual learning structure that maintains spatial information across numerous convolutional layers by incorporating residual skip connections. These skip connections help solve the vanishing gradient problem and facilitate the stable extraction of the hierarchical structure features without much information loss. These attributes are especially useful in the case of damage classification for downstream satellite-based building analysis, where details of roof architecture, geometric patterns, and structural boundaries are vital.

One pre-disaster building crop for each resolution crop (224×224 pixels) was fed into the ResNet50 feature extraction pipeline. The network then generates a final global average pooling, which results in a 2,048-dimensional feature embedding of the spatial and structural properties of the intact building. This embedding is the basis for the structural descriptor in Fusion+ (baseline embedding).

All convolutional parameters are kept fixed (`requires_grad = False`) to maintain the generalization of features, which are eventually used in Stream A to improve the computation efficiency. The motivation of this transfer learning strategy is based on two factors. First, low-level and mid-level structural properties already present in the images of ImageNet are well represented and are transferable to remote sensing images. Second, freezing the backbone causes catastrophic forgetting to be avoided, while significantly decreasing training complexity, GPU memory usage and convergence time.

3.3 Stream B: Post-Disaster Textural Debris Analysis (EfficientNet-B0)

The objective of Stream B is to learn fine-grained textual and surface-level damage representations from post-disaster satellite images with an efficient pre-trained backbone model, EfficientNet-B0. This stream is oriented towards visual abnormalities that occur locally as a result of an event associated with a disaster, such as debris, collapse of roofs, burn scars, flood marks, discoloration of materials and surfaces, etc. as opposed to the stream A which primarily emphasized the structural geometry.

The efficientNet-B0 model was chosen because it includes a compound scaling approach that uniformly scales the depth, width and input resolution of the network to obtain a strong representational performance at a relatively low computational cost. This architecture is very useful in obtaining high-frequency texture and irregular spatial patterns that frequently occur in places affected by disasters. The network's multi-scale feature extraction feature allows it to detect minor inconsistencies in the visual features that might not be adequately captured by a structural analysis.

Resolution 224×224-pixel building crops are individually preprocessed through the EfficientNet-B0 layers of feature extraction and then further processed using an adaptive average pooling operation. The output is then converted to a small feature vector of 1,280 dimensions that describes the textural and surface properties of the damaged structure. This embedding is used for downstream fusion and classification and complements the structural representation extracted by Stream A, which contains information specific to the disaster.

The parameters of the convolutional layers are frozen during training in Stream B, like in ResNet50 branch. By freezing the pretrained EfficientNet-B0 backbone, general low-level feature detectors that were learned from ImageNet are preserved, while at the same time computational overhead, memory and training complexity is significantly reduced. The trainable burden is thus completely moved to the light-weight fusion and classification head, which facilitates efficient model convergence and practical deployment on limited GPU resources.

3.4 Feature Fusion Layer and MLP Classification Head

After feature extraction, the 2,048-dimensional structural embedding from Stream A and the 1,280-dimensional textural embedding from Stream B are then concatenated together along the feature dimension to form a joint 3,328-dimensional representation. The fusion process can be represented as:

$$f_fused = [f_A \parallel f_B] \in \mathbb{R}^{3328}$$

This feature concatenation strategy is able to maintain all representational information of the two different streams without the information compression or weighting required in the additive or attention-based fusion methods. The model, therefore, preserves structural geometry information from pre-disaster images and textural information from a post-disaster image that is peculiar to a disaster.

The fused 3,328-dimensional feature vector is then fed into a lightweight multi-layer perceptron (MLP) classification head that is trained to learn cross-domain feature interactions and make four-class damage

prediction. The classification module consists of the following layers: a bottleneck linear transformation down to 512 neurons, Batch Normalization, ReLU activation and dropout regularisation with 0.4 dropout probability. This is a bottleneck layer that promotes small discriminative representation learning and lowers the number of features and spatial noise.

The intermediate representation goes through a second fully connected layer of size 512→256 which is followed by Batch Normalization, ReLU activation, and 0.3 dropout followed by a third transformation layer of size 256→128 with Batch Normalization and ReLU nonlinearity. Finally, a linear output layer with 128 units and Softmax activation is added to produce a probability distribution over the 4 damage classes of the disaster: No Damage, Minor Damage, Major Damage, and Destroyed.

The addition of Batch Normalization layers helps stabilize the flow of gradients and speed up the convergence of learning; the progressive decrease in dropout regularization decreases the co-adaptation of neurons, and thus enhances the generalization performance for the balanced training set. Overall classification head is a lightweight computational component that allows for efficient training without compromising the discriminative power.

4. Experimental Setup and Training Dynamics

4.1 Dataset Engineering and Class Balancing

The experimental evaluation is based on the xBD Dataset (xView2) benchmark dataset, which is made up of 2,799 pre-disaster and post-disaster satellite images from a number of large-scale natural disasters from around the world (hurricanes, floods, wildfires, tsunamis, earthquakes, volcanic eruptions). The dataset includes detailed polygon-level building annotations, and a universal damage classification scheme with four categories: No Damage, Minor Damage, Major Damage and Destroyed. The raw data is very heavily skewed with 75% of the building instances marked as No Damage, and just 7% marked as Destroyed. This imbalance will heavily influence deep learning models, causing them to make high-class predictions and thereby making them less sensitive towards very damaged structures.

Hence, a dedicated preprocessing and dataset engineering pipeline was developed to tackle this issue. The building-centric image crops were extracted, using Well-Known Text (WKT) representations, from the annotation JSON files. Every polygon was transformed to a bounding box (with contextual padding of 5 pixels around the polygon). The corresponding pre and post disaster image regions were then extracted and resized to a common resolution of 224×224 pixels. The paired crops were horizontally stitched together to create a single 448 x 224 composite image for efficient storage and retrieval.

In order to reduce class imbalance, a controlled extraction strategy was used where the number of samples extracted was stopped and the number of 10,000 was reached for each class. We got a dataset with 40,000 samples. The dataset has a number of samples in each damage class of the dataset. We split the dataset into two parts. 80 Percent of the dataset is for training. This is 32,000 samples of the

dataset. Then 20 Percent of the dataset is for validation. This is 8,000 samples of the dataset. We set up DataLoaders for the dataset, with a batch size of 32 and two worker threads to train the dataset faster.

4.2 Optimization and Regularization Strategy

The training strategy uses a transfer learning-based optimization scheme which only optimizes the parameters of the classification head, while the parameters of backbone feature extractors are frozen. The Adam optimizer is used for the optimization, with an initial learning rate of 0.001 and an L2 weight decay of 1×10^{-4} . This setup is a compromise between convergence speed and regularization, which helps to prevent overfitting on the relatively small fused feature space. The learning objective is formulated in terms of the standard Cross-Entropy Loss function of four classes of disaster damage.

A ReduceLROnPlateau scheduler is used to enhance the stability of training and dynamically adjust the learning rate. The scheduler takes care of checking the loss of validation set and after 2 epochs of tasting for too long (patience = 2), it decreases the learning rate by a factor of 0.5. This adaptation method can make the model jump out of suboptimal local minimums and improve the convergence process in the latter training.

Furthermore, early stopping is used as one of the most important regularization methods to avoid overfitting. The monitoring measure is the Validation Macro F1-Score because it is robust in the face of class imbalance. If there is no validation F1 improvement for the last 5 epochs, training is stopped (patience = 5). Checkpointing the best performing model weights, based on highest validation Macro F1, during training and restoring those weights after convergence.

While a maximum training epoch of 50 is set, the early stopping is usually triggered at an earlier epoch, with this study achieving a stable level of generalization performance at epoch 29 without the need to over-train.

5. Quantitative Evaluation and Analytics

5.1 Performance Metrics Framework

For disaster evaluation, the performance metrics should be able to deal with multi-class prediction complexity and severe class imbalance. The Macro-averaged F1-Score is used as the main evaluation measure in this study, which calculates the precision and recall values for each damage category and then averages the F1-Scores of these categories, giving equal influence on the average. Macro F1 does not rely on the proportion of samples to evaluate the performance of the model as overall accuracy does, instead it gives a more balanced assessment of the robustness of the model, especially for important classes that are not well represented, like Major Damage or Destroyed.

In the context of disaster assessment, the Macro F1 metric is significant as a model that is very accurate but misses the severely damaged class may get a high rate of accuracy, and still not detect the case damage. Since the minority classes are also considered, Macro F1 directly penalizes misclassification of

rare but high priority classes of disasters and thus is better adapted for humanitarian response applications.

Several supplementary evaluation metrics are also used, in addition to Macro F1, to gain a thorough understanding of classification behavior. These include per-class precision, recall, and F1 Score giving detailed information about the quality of prediction per class and the confusion between neighbouring damage levels. The overall classification accuracy is also provided as a general performance indicator. In addition, the confusion matrix analysis is performed to provide a visual representation of the inter-class misclassification pattern and highlight areas of feature overlapping which may occur especially between the two categories Minor Damage and Major Damage that are similar in visual appearance.

5.2 Convergence Analysis

The training dynamics of the proposed Fusion+ framework are stable and effectively converging in the optimization process. A maximum of 50 epochs were used for training, with the early stopping criterion stopping optimization at epoch 29 when the validation Macro F1-Score did not improve over the past 5 consecutive epochs. This is a good sign that the model has reached the optimum generalization performance before the maximum number of epochs without overfitting.

Note that the training loss decreases monotonously over the epochs from 0.8800 at epoch 1 to 0.3884 at epoch 29, showing that the trainable classification head was able to learn discriminative representations from the fused structural and textural feature embeddings during training, while being used on the frozen backbone extractors. The loss reduction is gradual, which indicates the stable gradient flow and the efficient parameter optimization in the lightweight MLP classification module.

The validation loss, however, shows moderate oscillating behavior at the end of training, a behavior typical of transfer learning systems in which only a fairly small number of parameters can be modified. This variability could also be due to differences in mini-batch composition and the inherent visual ambiguity between the intermediate damage categories. However, the slight changes of the validation Macro F1-Score in the training phase are evident and seem to be increasing over time from epoch 1 (0.7084) and reaching a peak before convergence (0.7669).

As the training loss keeps decreasing while the validation loss continues to oscillate, it is indicative that the model keeps improving the class-discriminative boundary but retains good generalization performance. Overall, the convergence behavior suggests that the optimization and regularization techniques proposed are effective in achieving a good trade-off between learning efficiency and the prevention of overfitting in the Fusion+ architecture.

Table 1. Selected Training Epoch Metrics (Key Checkpoints)

Epoch	Train Loss	Val Loss	Val F1	Note
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01	0.8800	0.7677	0.7084	Initial baseline
05	0.7046	0.7800	0.7353	Steady improvement
09	0.6171	0.6570	0.7543	LR begins to decay
14	0.5331	0.6443	0.7630	New best F1 checkpoint
17	0.4719	0.7684	0.7657	Best F1 = 0.7669
24	0.4093	0.8435	0.7669	Best model saved
29	0.3884	1.4438	0.7624	Early stopping triggered

Selected epochs from the 29-epoch training run. Early stopping triggered at epoch 29 after 5 epochs without F1 improvement.

5.3 Granular Confusion Matrix Assessment

The classification report in Table 2 is obtained after final evaluation on the 8,000 samples of the validation set. The model has an overall accuracy of 77% and Macro F1-Score of 0.77.

Table 2. Final Classification Report — Fusion+ Dual-CNN Framework

Class	Precision	Recall	F1-Score	Support
No Damage	0.82	0.81	0.82	2047
Minor Damage	0.70	0.76	0.73	1991
Major Damage	0.70	0.69	0.70	1984
Destroyed	0.85	0.81	0.83	1978
Accuracy			0.77	8000
Macro Avg	0.77	0.77	0.77	8000
Weighted Avg	0.77	0.77	0.77	8000

Per-class precision, recall, F1-score and support on the 8,000-sample validation set.

The detailed confusion matrix analysis highlights interesting behaviour of the proposed Fusion+ framework for each class. The Destroyed class has the best precision of 0.85 and F1-score of 0.83 among all categories. This finding suggests that when a building is severely collapsed, visual patterns are very distinctive and can be easily identified even from overhead satellite images. These are all strong discriminative features that are captured well by the dual-stream and are extensively observed in the form of large debris fields, roof disappearance and fragmented building footprints, along with substantial surface disruption.

Similarly, No Damage class has a good classification ability, with an F1 score of 0.82. In these cases, the baseline spatial representation of these image pairs of pre-and post-disaster images is not significantly different, and the model is capable of sustaining the spatial representation in a reliable manner and accurately identifying intact buildings.

The largest classification challenge, however, is between the Minor Damage and Major Damage classes, where the two classes have the lowest F1 scores of 0.73 and 0.70 respectively. A look at the confusion matrix shows that 309 Minor Damage samples are classified as Minor Damage and 366 Major Damage samples are classified as Major Damage. This error mode is bi-directional confusion the most dominant error mode.

The results clearly confirm the original hypothesis that the intermediate levels of structural damage have intrinsic visual similarities when viewed from a satellite nadir viewing angle. Many subtle features like partial roof deformation, disruption of the facade at the local level, small amounts of debris and minor wall cracking are often close to the boundary separation resolution of satellite imagery, making separation very difficult. As a result, the features' distribution within these neighbouring categories is very correlated, leading to more ambiguity in decisions for classification.

6. Qualitative Discussion and Visual Case Studies

6.1 Visual Inference Testing Results

The real world inference capability of the proposed Fusion+ framework was assessed through three qualitative case studies with pre-disaster and post-disaster image pairs not present in the validation subset, selected from geographically different disaster events. The model is given paired building crops as input and a predicted damage category and a confidence score as output for each sample using the Softmax probability estimation method. These visual inference experiments shed light on the interpretability, robustness, and decision behavior of the dual-stream architecture for realistic disaster situations.

The successful classification of the No Damage category is shown in Case Study 1, which has a confidence score of 73.3%. The consistency of the building crop before and after the disaster is good, as evidenced by the visual inspections of the roofline geometry, alignment of the structural footprint and limited variation at the surface level. The prediction is right, but the confidence score is still reasonably low, since it is often hard to reliably determine that there is no subtle structural damage, from satellite images.

The model achieves a high confidence score of 88.9% for predicting Major Damage in Case Study 2. The post disaster image shows significant structural deformation combined with non-uniform debris distribution and some visible surface disturbance compared to the pre-disaster image. It can be observed that the strong confidence level means that the Fusion+ framework successfully combines structural elements of Stream A and post-disaster textural degradation elements of Stream B.

Case Study 3 is one of the most difficult examples in the data set, with a relatively low prediction confidence score of 45.3% for Minor Damage. The low-confidence prediction implies that the learned decision boundary has similar probability distributions for neighboring damage categories, such as Minor Damage and Major Damage. This behaviour is in line with the confusion matrix analysis, and

reflects the real visual level of ambiguity observed through orbital satellite images between the different levels of structural damage.

6.2 Limitations and Future Horizons

Although the overall effectiveness of classification is good, there are a number of shortcomings in the existing Fusion+ framework. First, the model only uses imagery from the top down (nadir) satellites, so important indicators of structural damage, like wall collapse, window failure, facade deformation, and foundation cracking are not included. Several of these characteristics are more easily seen from the oblique or multi-angle perspectives. Multi-view remote sensing fusion strategies which can include non-nadir view observations and improve structural interpretability/damage localization should therefore be investigated for future research.

Secondly, the frozen-backbone transfer learning approach, while being cost-effective and less complex to train, restricts the ability of the feature extractors to learn the satellite-specific visual features. The image-net-trained networks may not be well-trained for the spectral or geometric features that are typical of remote sensing datasets. In future, one will consider learning rates that adapt slowly to fine-tune the backbone to better fit the domain of interest while keeping training stable with discriminative learning rates.

The other is the lack of the modelling of broader spatial context. The existing architecture takes a building-centric approach to classification without taking into account other buildings in the vicinity or the environment. But in an urban area, disaster damage is often spatially correlated within cities. Contextual reasoning mechanisms like attention-based spatial modeling, graph neural networks (GNNs), and transformers at the scene level could enhance the classification's robustness in high-density environments.

A number of future directions are proposed for further development of the Fusion+ framework. These involve using Vision Transformer (ViT)-based networks for improved global feature representation, replacing the standard MLP classification head with a Kolmogorov-Arnold Network (KAN) and applying synthetic augmentation on rare classes of disaster categories using Generative Adversarial Networks (GANs) or diffusion-based text-to-image models. These improvements might further boost the models' generalization, scalability, and readiness for humanitarian deployment in the real world.

7. Conclusion

This paper introduced Fusion+ -- a dual-stream deep learning framework to automate the four-level building damage classification based on paired pre-disaster and post-disaster satellite imagery. The proposed architecture is a fusion of two pre-trained CNN backbones: ResNet-50 for learning structural and geometric features from pre-disaster images, and EfficientNet-B0 for post-disaster image degradation patterns, including debris fields, roof collapse and surface anomalies. To do a 4-class

disaster damage prediction, the extracted feature embeddings are concatenated together and fed to a lightweight bottleneck multi-layer perceptron classification head for fusion.

A balanced subset of xBD Dataset benchmark dataset was extracted consisting of 40,000 image pairs of buildings, equally distributed across the categories of No Damage, Minor Damage, Major Damage and Destroyed, to serve as the experimental set. The proposed framework achieved a satisfactory Macro F1-Score of 0.77 on an 8,000 sample validation set with strong and balanced classification performance for both majority and minority (damage) classes. The model showed high discriminative capability for the Destroyed and No Damage classes because of the different structural characteristics that stand out in the image.

The results also show that the lightweight transfer learning approach using the frozen feature extractor is an efficient but research-competitive approach for satellite-based disaster assessment. The architecture greatly reduces the complexity of training, GPU memory consumption and convergence time by freezing the pretrained backbone networks and optimizing only the classification head, without sacrificing the backbone networks' feature generalization capability. Moreover, for the practical disaster datasets, the proposed class-balanced extraction method can effectively reduce the severe long-tail imbalance without any use of synthetic sample generation techniques. Early stopping and Macro F1 selection of checkpoints further guaranteed stable convergence and optimal validation performance.

The confusion matrix analysis showed that the most common mistake was between the Minor Damage and Major Damage classes, which are all very similar in terms of the structure of the objects, and close to the spatial resolution limit of the nadir satellite imagery. The limitation is not a modeling issue only but a fundamental problem with overhead remote sensing and a promising line of future research to consider with contextual reasoning, transformer based architectures, and multi-angle image fusion.

Overall, the Fusion+ framework has promising potential for use as a scalable, operationally deployable decision-support system for humanitarian disaster response. The architecture is proposed for rapid and automated building damage triage from satellite images, to assist emergency response and relief efforts in time-critical applications post-disaster.

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